



AGENT SYSTEMS

Autonomous Multi-Agent Architecture

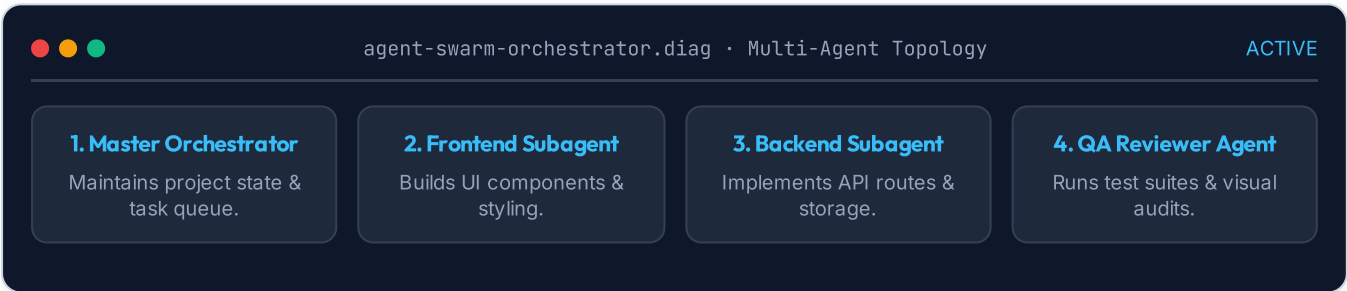
Engineering high-throughput collaborative AI agent swarms and hierarchical task orchestration.

EXECUTIVE SUMMARY

Scaling AI automation beyond single-turn prompts requires dividing responsibilities across specialized agents. This guide outlines orchestrator-worker topologies, shared context synchronization, and quota-aware execution boundaries.

1 Master Orchestrator & Worker Topology

A central orchestrator agent evaluates the high-level goal, generates an atomic task breakdown, and spawns dedicated subagents in parallel. Subagents report status back to the coordinator while maintaining isolated working context.



2 Production Implementation & Schema

```
// Parallel Subagent Dispatch Loop (Node.js)
async function dispatchSwarm(tasks) {
  const results = await Promise.allSettled(
    tasks.map(task => spawnAgent({ role: task.role, spec: task.prompt }))
  );
  return results.map((r, i) => ({ task: tasks[i].id, status: r.status }));
}
```



3 Shared Context & State Synchronization

Agent swarms communicate through lightweight state files (STATE.json) and shared vector memory, preventing memory leaks and avoiding context degradation during long refactoring runs.

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swarm-monitor • Multi-Agent Status

CONNECTED

```
[ORCHESTRATOR] Assigned task "build-api" to Subagent-1 (PID: 3291)
[ORCHESTRATOR] Assigned task "build-ui" to Subagent-2 (PID: 3292)
[SUBAGENT-1] API endpoint created • HTTP 200 verification pass
[SUBAGENT-2] UI components rendered • Screenshot captured
[MERGE] Task complete • 0 merge conflicts detected
```

4 Key Takeaways & Best Practices

✓ Context Isolation

Keep subagent prompts focused on single tasks to prevent hallucination and maintain high reasoning quality.

✓ State Serialization

Persist intermediate progress in JSON files so disconnected or restarted sessions pick up without data loss.

✓ Quota Management

Set hard boundaries on token consumption per subagent to avoid unexpected API budget overruns.

✓ Deterministic Merges

Require subagents to run automated test suites before committing changes back to the main branch.

IMPLEMENTATION SUPPORT

Need engineering assistance integrating these architectures into your business? Contact our team directly at futureai4all@gmail.com.